Charging Forward

EV Infrastructure Analytics for Charlottesville

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Executive Summary

EV charging demand is stable with seasonal spikes, and infrastructure decisions should prioritize high-utilization locations while maintaining connector flexibility.

Project Scope

- Cleaned and analyzed Charlottesville EV charging station data
- Built a cloud-based ETL pipeline using Google Cloud Storage, Python, and SQLite
- Integrated historical weather data using the Open-Meteo API

Key Findings

- Demand is stable with short-term spikes
- Usage increases in moderate temperatures
- Rainfall has negligible impact
- No clear dominant connector trend
- A small set of stations drives highest usage

Recommendation

- Prioritize high-utilization stations
- Maintain balanced connector strategy with CCS emphasis

Cloud Pipeline Architecture

Deliverables Demonstrated:

- Authenticated access to GCS bucket
- Automated data integration and upload pipeline
- Cleaned outputs stored in team cloud folder
- Reproducible notebook environment

Google Cloud Storage Bucket (raw municipal datasets)

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Google Colab + Python ETL

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Cleaned Files + SQLite Outputs

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Data Analysis + Visualizations

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Strategic Recommendation

```python
from google.colab import auth
auth.authenticate_user()

from google.cloud import storage
client = storage.Client(project="ds2002-492012")
bucket = client.bucket("ds2002-capstone-sp26-v2")

# List files in your team folder
blobs = list(bucket.list_blobs(prefix="team-16/"))
for b in blobs:
    print(b.name)
```

```
team-16/.keep
team-16/Test Doc.docx
team-16/Test_Upload.txt
team-16/cleaned_operators.csv
team-16/cleaned_sessions.csv
team-16/cleaned_stations.csv
team-16/cleaned_vehicles.csv
team-16/ev_analytics.db
```

Working with GCS introduced challenges around authentication and file paths, particularly when debugging access issues. Compared to local files, cloud storage required more setup but improved collaboration and ensured consistent data across the team.

Turning Raw Data into Reliable Data

Key Cleaning Steps

- Removed duplicates and invalid records
- Fixed timestamps and removed negative kWh
- Standardized vehicle IDs (VH-XXX format)
- Standardized station IDs

Why It Matters

- Raw data contained inconsistent formats and duplicate identifiers
- Cleaning created a reliable foundation for analysis

Vehicle ID fragmentation inflated the number of unique vehicles from 14 to over 40 categories. Standardizing IDs significantly changed the interpretation of demand patterns. Without this step, results would have been misleading and overestimated vehicle diversity.

Turning Raw Data into Reliable Data

Data Quality Issue	Example	Resolution
Duplicate charging sessions	repeated records	removed
Vehicle ID fragmentation	Tesla M3 / Model 3 / TESLA 3	consolidated
Negative energy values	-14.2 kWh	excluded
Inconsistent dates	mixed formats	standardized
Station ID mismatch	ST01 / ST-01	unified

Without cleaning, station demand, vehicle share, and connector trends would have been materially misstated.

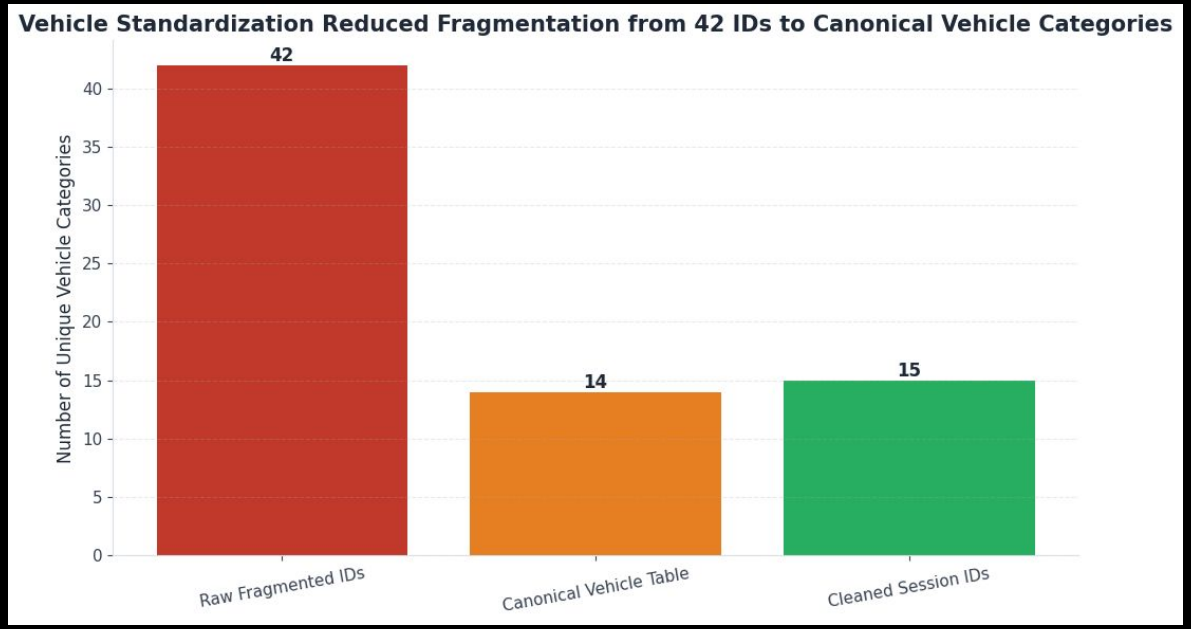
Vehicle Standardization Impact

Key Takeaway:

Cleaning reduced vehicle categories from 42 → ~14

Insight:

- Eliminated artificial fragmentation
- Improved analytical accuracy



Reflection:

We chose to standardize inconsistent IDs instead of dropping them. While removing unclear entries would simplify the dataset, it would also reduce sample size and introduce bias. Standardization preserved data while improving reliability.

Charging Demand Over Time

Key Takeaway:

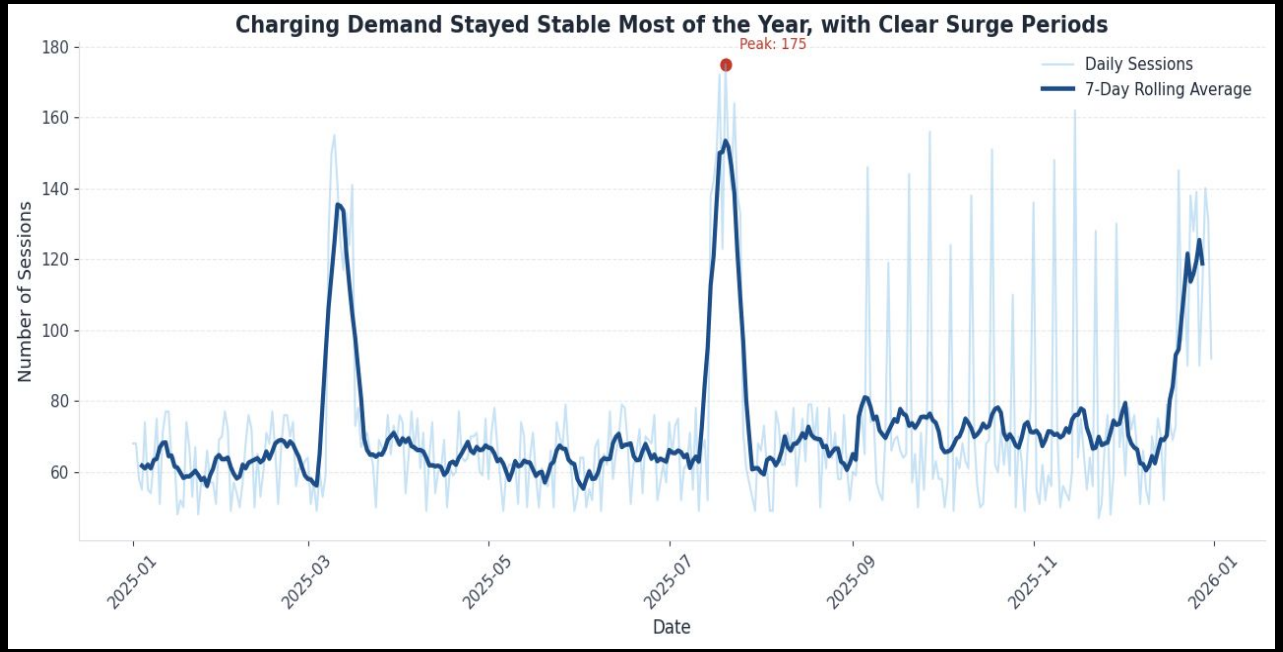
Charging demand is stable overall, with clear seasonal spikes

What the Data Shows:

- Baseline ~60-80 sessions/day
- Peaks in March, July, and December

Insight:

Demand varies by season, not long-term growth



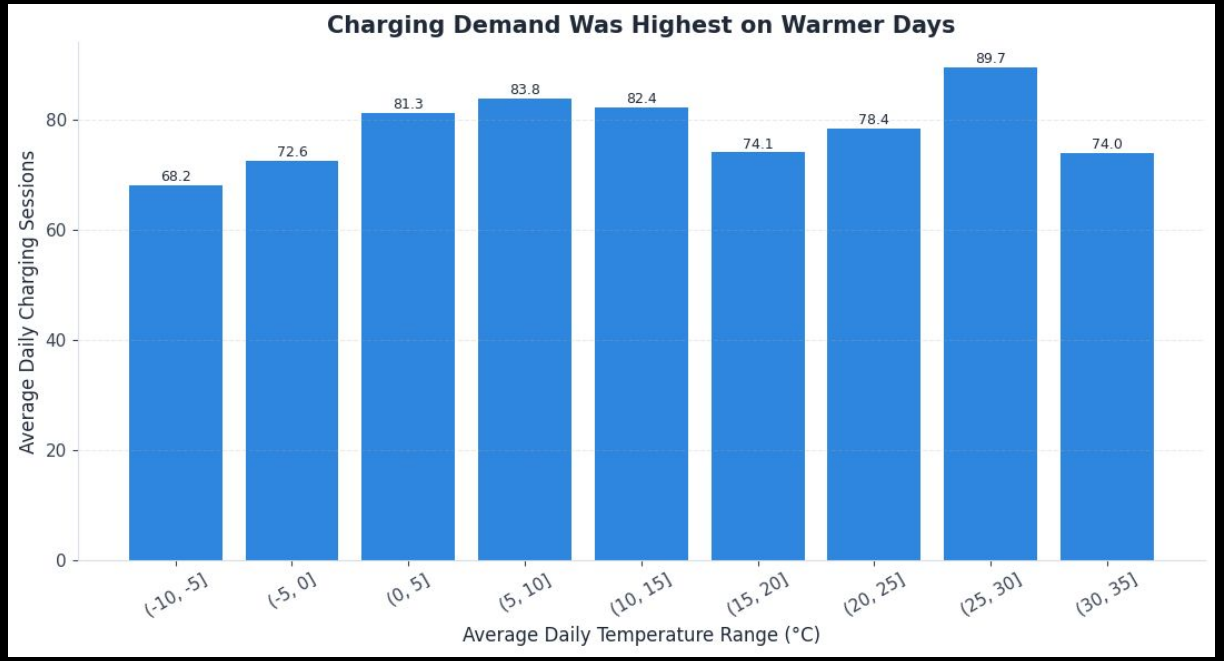
Temperature & Charging (API)

Key Takeaway:

Charging demand is higher in moderate-to-warm temperatures

What the Data Shows:

- Lower demand in colder temperatures
- Peak usage around 20-30°C
- Relationship is not strictly linear



Reflection:

This analysis depends on external API data, making it a fragile part of the pipeline. API downtime or formatting changes could break the workflow, emphasizing the need for validation and error handling.

Rainfall Impact (API)

Key Takeaway:

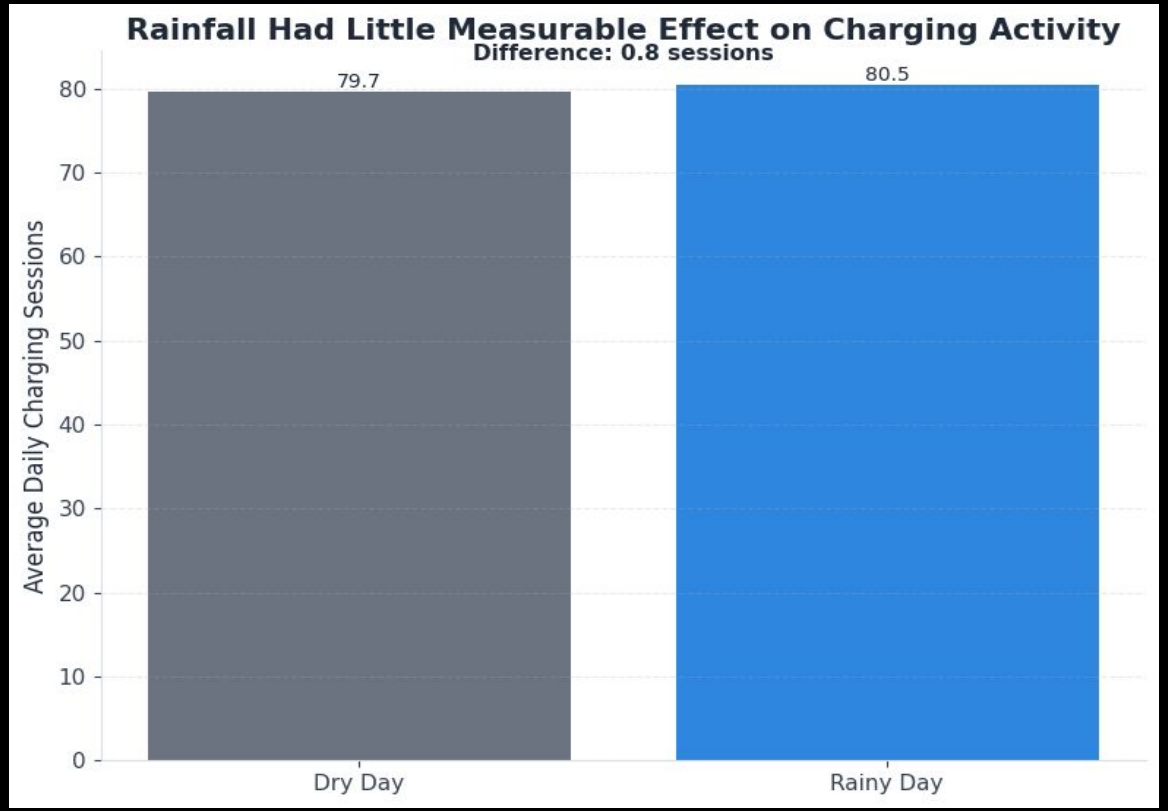
Rainfall had negligible impact on charging activity

What the Data Shows:

- Less than 1 session difference between rainy and dry days

Insight:

Charging behavior is consistent regardless of short-term weather



Connector Trends

Key Takeaway:

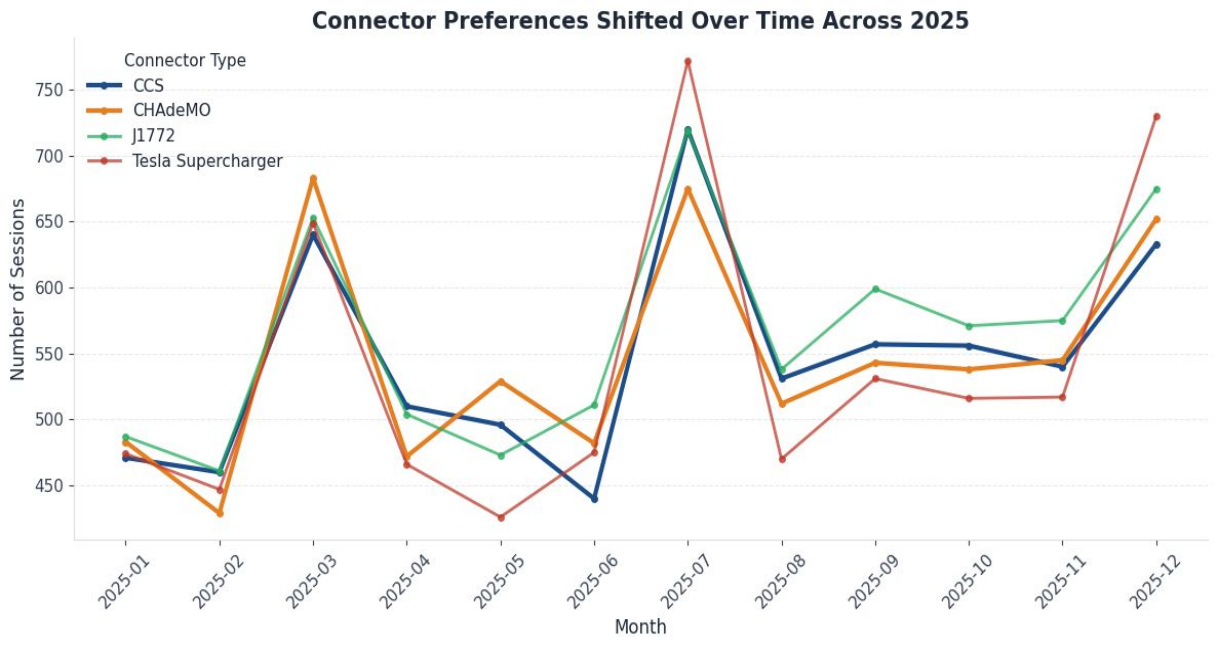
Connector usage followed similar seasonal patterns with no clear dominant trend

What the Data Shows:

- All connector types move together
- No strong divergence between CCS and CHAdeMO

Insight:

Current demand is distributed across connector types



Top Stations

Key Takeaway:

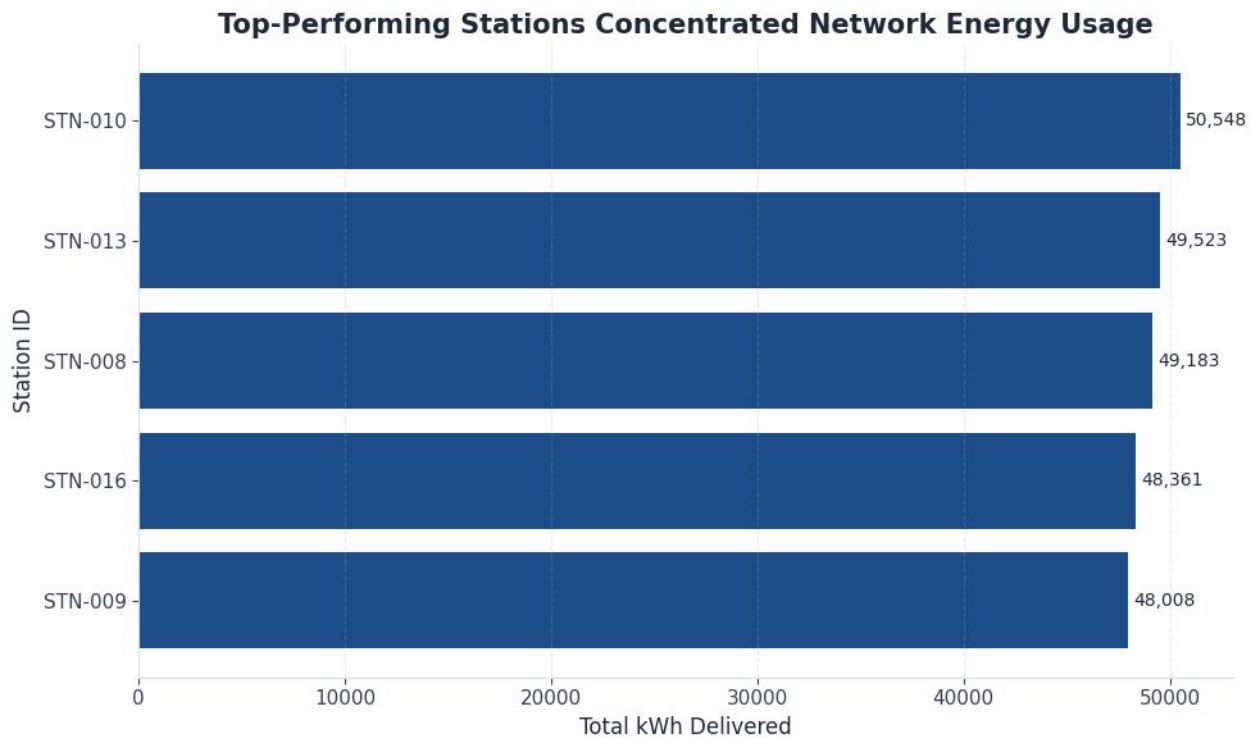
A small group of stations consistently delivers the highest energy usage

What the Data Shows:

- Top stations cluster around similar high output

Insight:

Infrastructure planning should prioritize high-utilization locations



Why CCS Supports Long-Term Scalability

Current Data Context

- Our data does not show a clear dominant connector trend.
- Connector types follow similar seasonal demand patterns.

Why CCS Still Matters

- CCS is defined in federal NEVI standards as a standard interface for DC fast charging.
- NEVI-funded DC fast-charging ports must be capable of charging CCS-compliant vehicles.
- Corridor-serving DC fast chargers must support at least 150 kW per port, making CCS relevant for scalable fast-charging expansion

Strategic Takeaway

- Maintain connector flexibility today.
- Prioritize CCS-compatible upgrades for future DC fast-charging investment.

Source: Federal Highway Administration / eCFR, 23 CFR Part 680.

Recommendation: Prioritize High-Utilization Stations + CCS Expansion

Based on observed demand trends and connector adoption patterns, future capital investment should focus on prioritizing high utilization stations and expand CCS charger deployment.

Supporting Reasons:

- Demand is concentrated geographically
- No clear connector dominance in current data
- CCS aligns with broader industry trends

Alternative Considered → Maintain equal investment across all connector types.

Why Rejected → Does not optimize for usage or future scalability

Final Takeaways

Infrastructure Insights

- Demand is stable but concentrated
- Expansion should focus on high-usage stations

Connector Strategy

- No dominant connector in current data
- CCS prioritized for long-term scalability

Data & Pipeline

- Data cleaning was essential for valid analysis
- Cloud pipelines improved reproducibility
- API integration added context but increased fragility

Our team divided responsibilities across pipeline development, data cleaning, API integration, and analysis. This allowed efficient parallel work while maintaining consistency through shared datasets. With more time, we would expand analysis and refine visualizations, but collaboration was key to building a complete pipeline.